

SE 216 – SOFTWARE PROJECT MANAGEMENT
PROJECT RISKS DOCUMENT

PROJECT NAME: REPLATE

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LIKELIHOOD RANK	RISK DESCRIPTION
1	Dependency on external services: APIs such as Google Maps or notification services are highly likely to create issues such as rate limits, downtime, or unstable responses during development and testing.
2	Testing and integration difficulties: Multi-role workflows, QR verification, and real-time operations make testing highly complex and likely to produce unexpected bugs.
3	Integration problems between system components: Backend, database, and external APIs may not work smoothly together during implementation.
4	Communication problems within the team: Miscommunication may occur during task distribution and collaborative development.
5	Real-time synchronization problems: Reservation updates may not synchronize correctly between users, businesses, and municipal distribution centers during concurrent operations, causing inconsistent stock or reservation information.
6	Time management issues: Tasks may take longer than expected, especially during integration and debugging phases.

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7	System performance issues under concurrent operations: Real-time updates and map-based operations may reduce responsiveness during stress testing.
8	Version control conflicts: Merge conflicts and overwritten code may occur when multiple developers work on the same modules.
9	Lack of experience with selected technologies: Team members may initially struggle with some frameworks and tools, but this can improve over time.
10	Team member unavailability or low contribution: A member becoming unavailable is possible but relatively less likely in the current team structure.

IMPACT RANK	RISK DESCRIPTION
1	Integration problems between system components: Failure in integration may block major functionalities and delay the entire project significantly.
2	Real-time synchronization problems: Synchronization failures may cause incorrect reservation and stock information, directly affecting the core functionality of the system.
3	System performance issues under concurrent operations: Poor performance may negatively affect reservations, map rendering, and real-time updates.
4	Testing and integration difficulties: Undetected bugs in QR verification, reservation workflows, or multi-role operations may destabilize the system.
5	Dependency on external services: External API failures may temporarily disable important features such as maps and notifications.

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6	Team member unavailability: Losing contribution from a member may significantly affect workload distribution and development speed.
7	Time management issues: Delays may reduce the completeness and quality of the final product.
8	Lack of experience with technologies: Inexperience may slow development and lead to inefficient implementations.
9	Communication problems within the team: Miscommunication may create inconsistencies and reduce team efficiency.
10	Version control conflicts: Git conflicts may slow development but are generally manageable.

LIKELIHOOD RANK	IMPACT RANK	COMBINED RANK	RISK DESCRIPTION
3	1	4	Integration problems between system components
1	5	6	Dependency on external services
2	4	6	Testing and integration difficulties
5	2	7	Real-time synchronization problems
7	3	10	System performance issues under concurrent operations
6	7	13	Time management issues
4	9	13	Communication problems within the team
10	6	16	Team member unavailability or low contribution
9	8	17	Lack of experience with selected technologies
8	10	18	Version control conflicts